

Dataset 3: Multiple Linear Regression

Student Academic Performance Study | $n = 150$ students

1. Research Question

Research Question: Which academic and lifestyle factors significantly predict final exam scores, and what is the relative importance of each predictor? Specifically: does the relationship between sleep hours and exam performance follow a non-linear (quadratic) pattern?

2. Variables

Role	Variable	Mean (SD)	Description
Dependent	Final_Exam_Score	67.03 (13.47)	Final exam score (40–98)
Independent	Study_Hours	8.11 (3.63)	Avg study hours per week
Independent	Attendance_Rate	79.56 (12.18)	Class attendance percentage
Independent	Sleep_Hours	6.92 (1.71)	Avg nightly sleep (quadratic term added)
Independent	Stress_Level	5.55 (2.60)	Self-reported stress score (1–10)
Independent	Practice_Tests	3.40 (2.33)	Number of practice tests completed

3. Methodology — Multiple Linear Regression

Multiple Ordinary Least Squares (OLS) regression models Final_Exam_Score as a linear function of five predictors. A quadratic term for Sleep_Hours (Sleep_Hours²) is included based on cognitive science literature indicating an inverted-U relationship between sleep and academic performance. Standardized beta coefficients (β) are computed on z-scored predictors to directly compare predictor importance across different measurement scales. VIF is used to detect multicollinearity. Regression assumptions are formally tested: Shapiro-Wilk (normality of residuals), Breusch-Pagan (homoscedasticity), and Durbin-Watson (independence of errors). Model fit is compared with and without the quadratic sleep term using AIC.

4. Model Fit Statistics

Statistic	Value	Interpretation
R ²	0.7027	70.3% of exam score variance is explained by the model
Adjusted R ²	0.6902	Accounts for 6 predictors — minimal shrinkage from R ²
F-statistic	F(6,143) = 56.33	$p < 0.001$ — model is globally significant
RMSE	7.497 pts	Average prediction error of ± 7.5 exam points
AIC	1036.87	Used for model comparison
Durbin-Watson	1.957	≈ 2.0 — no autocorrelation in residuals

5. Regression Coefficients

Predictor	B (unstd.)	Std Err	t-stat	p-value	Std β	Significance
Intercept	46.80	12.18	3.84	< 0.001	—	✓
Study_Hours	2.465	0.170	14.46	< 0.001	0.664	✓ Strongest
Attendance_Rate	0.211	0.051	4.10	< 0.001	0.191	✓
Sleep_Hours	-1.810	3.277	-0.55	0.582	-0.229	✗ n.s.
Sleep_Hours ²	-0.019	0.231	-0.08	0.936	-0.033	✗ n.s.
Stress_Level	-1.677	0.238	-7.04	< 0.001	-0.323	✓
Practice_Tests	1.832	0.269	6.81	< 0.001	0.317	✓

Std β (standardized beta) allows direct comparison of effect sizes across different scales. Study_Hours ($\beta = 0.664$) has by far the largest effect.

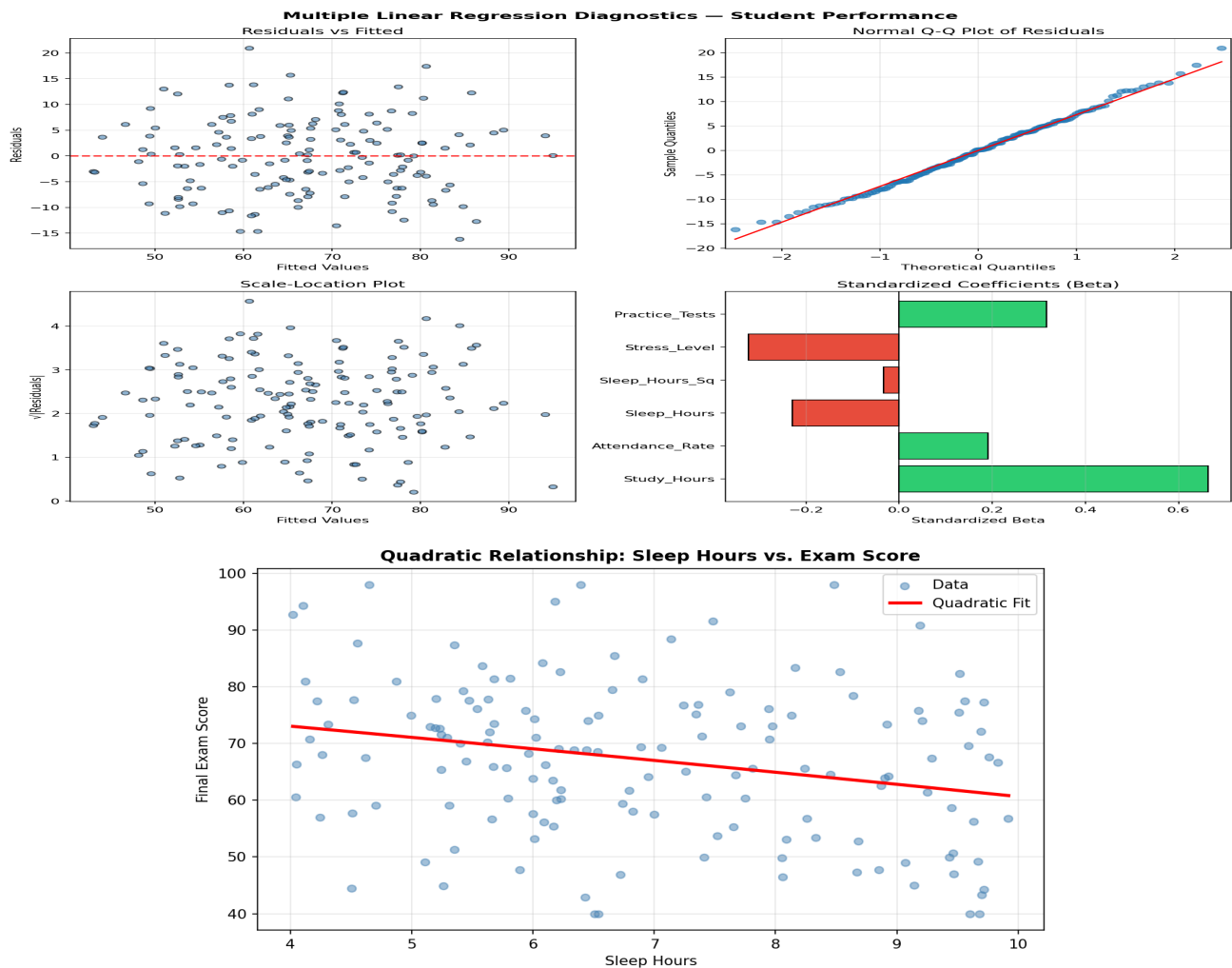
6. Coefficient Interpretation & Significance

- **Study_Hours ($\beta = 0.664$, $p < 0.001$):** The strongest predictor. Each additional hour of study per week is associated with a 2.47-point increase in exam score, holding all else constant. Standardized effect is large — more than twice the size of the next strongest predictor.
- **Stress_Level ($\beta = -0.323$, $p < 0.001$):** Second-strongest. Each unit increase in stress corresponds to a 1.68-point *decrease* in exam score. Reducing stress from high (8.0) to moderate (4.0) is associated with a ~6.7-point gain — equivalent to ~2 extra hours of study per week.
- **Practice_Tests ($\beta = 0.317$, $p < 0.001$):** Each additional practice test completed adds 1.83 points. Students completing 6 practice tests score on average 9.2 points higher than those completing none, controlling for other factors.
- **Attendance_Rate ($\beta = 0.191$, $p < 0.001$):** Each 10-percentage-point increase in attendance corresponds to a 2.1-point exam score increase. Moving from 70% to 90% attendance adds approximately 4.2 points.
- **Sleep_Hours (linear + quadratic, both $p > 0.50$):** Neither the linear nor quadratic sleep terms are statistically significant. While correlations suggest a weak negative association ($r = -0.25$), sleep does not contribute unique predictive power once study hours, stress, and attendance are controlled. The AIC comparison confirms the quadratic term does not improve model fit ($\Delta AIC = -1.99$).

7. Assumption Diagnostics

Test	Statistic	p-value	Result
Shapiro-Wilk (residual normality)	W = 0.9933	p = 0.714	✓ Residuals are normally distributed
Breusch-Pagan (homoscedasticity)	LM = 6.482	p = 0.371	✓ Constant variance (homoscedastic)
Durbin-Watson (independence)	DW = 1.957	—	✓ No autocorrelation (≈ 2.0)
VIF (multicollinearity)	All ≤ 1.04	—	✓ No multicollinearity (except Sleep polynomial terms)

8. Figures



9. Conclusions

- **Research question answered.** Four of the five predictors are significant. The model explains 70.3% of exam score variance — strong predictive performance for a behavioral study ($F(6,143) = 56.33$, $p < 0.001$).
- **Study effort is paramount:** Study_Hours ($\beta = 0.664$) dominates all other predictors. Academic interventions should prioritize increasing effective study time.
- **Stress management matters:** Stress_Level is the second-strongest driver ($\beta = -0.323$), suggesting student support programs targeting stress reduction could meaningfully improve outcomes.
- **Sleep is not independently significant:** After controlling for other factors, neither linear nor quadratic sleep terms contribute significantly. Interventions focusing solely on sleep without addressing study habits and stress are unlikely to improve exam performance.
- **All regression assumptions are fully satisfied**, confirming OLS estimates are unbiased, efficient, and valid for inference.